

02456 Deep Learning

Exercises Week 2

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Prince 5.9 Consider a multivariate regression problem in which we predict the height of a person in meters and their weight in kilos from data x . Here, the units take quite different ranges. What problems do you see this causing? Propose two solutions to these problems;

For a NN using MSE the loss will be defined as $L = (y_h - \hat{y}_h)^2 + (y_w - \hat{y}_w)^2$. The numerical difference from a guess with 10% error in height and error is big, and the weight error will dominate the loss function completely. To get rid of this problem we can standardize data to for example a scale between 0 and 1. We can also use a logarithmic error loss function instead such that we take the natural logarithm of every term in the MSE; $L = (\ln(1 + y_h) - \ln(1 + \hat{y}_h))^2 + (\ln(1 + y_w) - \ln(1 + \hat{y}_w))^2$. We add 1 to every term to prevent negative loss, and the case of $\ln(0) \approx -\infty$. Lastly we can measure the scalar difference in unit as for example: a 10% error in the two parameter outputs will give for height; $(1.8 - 1.62)^2 = 0.0324$ and for weight; $(80 - 72)^2 = 64$. If we fix the scale of weight $\lambda_w = 1$, then the scale of height $\lambda_h = \frac{64}{0.0324} \approx 1975$, such that $L = \lambda_h(y_h - \hat{y}_h)^2 + \lambda_w(y_w - \hat{y}_w)^2$ becomes a weighted term that accounts for the small numerical variance height has compared to weight.

Prince 6.5 Compute the derivatives of the least squares loss with respect to the ten parameters of the simple neural network model introduced in equation 3.1:

$$f[x, \phi] = \phi_0 + \phi_1 a[\theta_{10} + \theta_{11}x] + \phi_2 a[\theta_{20} + \theta_{21}x] + \phi_3 a[\theta_{30} + \theta_{31}x]$$

Think carefully about what the derivative of the ReLU function $a[\bullet]$ will be;

To find the derivatives of the inner parameters we first find all the outer partial derivatives such that we can formulate the parameter derivatives as products of these;

$$\frac{\partial L}{\partial f} = 2(f - y)$$

the next layer being the ϕ scalars;

$$\begin{aligned}\frac{\partial L}{\partial \phi_0} &= 2(f - y)(1) \\ \frac{\partial L}{\partial \phi_1} &= 2(f - y)(a[\theta_{10} + \theta_{11}x]) \\ \frac{\partial L}{\partial \phi_2} &= 2(f - y)(a[\theta_{20} + \theta_{21}x]) \\ \frac{\partial L}{\partial \phi_3} &= 2(f - y)(a[\theta_{30} + \theta_{31}x])\end{aligned}$$

now for the ReLU activation function a , we say $z = \theta_{10} + \theta_{11}x$ such that $a[z] = \max(0, z)$. If $z > 0$ then the derivative of $\frac{\partial a}{\partial z} = 1$ else it is 0. And then lastly we have $\frac{\partial z}{\partial \theta_{10}} = 1$ and $\frac{\partial z}{\partial \theta_{11}} = x$ (also true for other theta

pairs). This gives us;

$$\begin{aligned}\frac{\partial L}{\partial \theta_{10}} &= 2\phi_1(f - y) \text{ if } z > 0, \text{ else } 0 \\ \frac{\partial L}{\partial \theta_{11}} &= 2x\phi_1(f - y) \text{ if } z > 0, \text{ else } 0\end{aligned}$$

which again is the same for $\theta_{20}, \theta_{21}, \theta_{30}$ and θ_{31} .

Prince 7.10 Derive the equations for the backward pass of the backpropagation algorithm for a network that uses leaky ReLU activations, which are defined as:

$$a[z] = \text{ReLU}[z] = \begin{cases} \alpha \cdot z & z < 0, \\ z & z \geq 0 \end{cases}$$

where α is a small positive constant (typically 0.1);

The backpropagation algorithm propagates gradients backwards through network layers. Changing from ReLU to leaky ReLU only changes the computation of the loss gradient wrt. the layers re-activation z . We start by finding the partial derivatives of the output a wrt. the input z ;

$$\frac{\partial a}{\partial z} = I(z \geq 0) + \alpha I(z < 0)$$

with the indicator function returning 1 if the condition is met, and 0 if not. If we integrate that into the chain rule we get;

$$\frac{\partial L}{\partial z} = \frac{\partial L}{\partial a} \cdot (I(z \geq 0) + \alpha I(z < 0))$$

Exam 2025 Q3 Consider the following simple neural network with a single hidden layer:

$$h = \text{ReLU}(wx), \quad y = vh, \quad L = \frac{1}{2}(y - t)^2$$

where $x \in \mathbb{R}$ is the input, $h \in \mathbb{R}$ is the hidden layer output, $y \in \mathbb{R}$ is the output, $L \in \mathbb{R}$ is the loss, $t \in \mathbb{R}$ is the target, $w, v \in \mathbb{R}$ are weights, and $\text{ReLU}(z) = \max(0, z)$. Assume that $x = 1, t = 6, w = 2$, and $v = 4$. What is the value of the gradient of the loss $\frac{\partial L}{\partial w}$ for this input and target?

- A. 0
- B. 4
- C. 8
- D. 16

C. is the correct answer. We compute the gradient using the chain rule, where we need to work backward from the loss all the way to the weight;

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial y} \cdot \frac{\partial y}{\partial h} \cdot \frac{\partial h}{\partial z} \cdot \frac{\partial z}{\partial w}$$

where $z = wx$, the pre-activation of the hidden layer. We find these one by one; $\frac{\partial L}{\partial y} = \frac{1}{2} \cdot ((y - t)^2)' \cdot (y - t)' = \frac{1}{2} \cdot 2(y - t) \cdot 1 = y - t$ which for our values $y = 4 \cdot 2 = 8$ (since $h = \text{ReLU}(z) = 2$ for $z = 2 \cdot 1$) and $t = 6$ is $8 - 6 = 2$. $\frac{\partial y}{\partial h} = vh' = 4h' = 4$. $\frac{\partial h}{\partial z} = \text{ReLU}(z)' = \text{ReLU}(2)' = 1$. And lastly $\frac{\partial z}{\partial w} = wx' = 1w' = 1$. Giving us

$$\frac{\partial L}{\partial w} = 2 \cdot 4 \cdot 1 \cdot 1 = 8$$